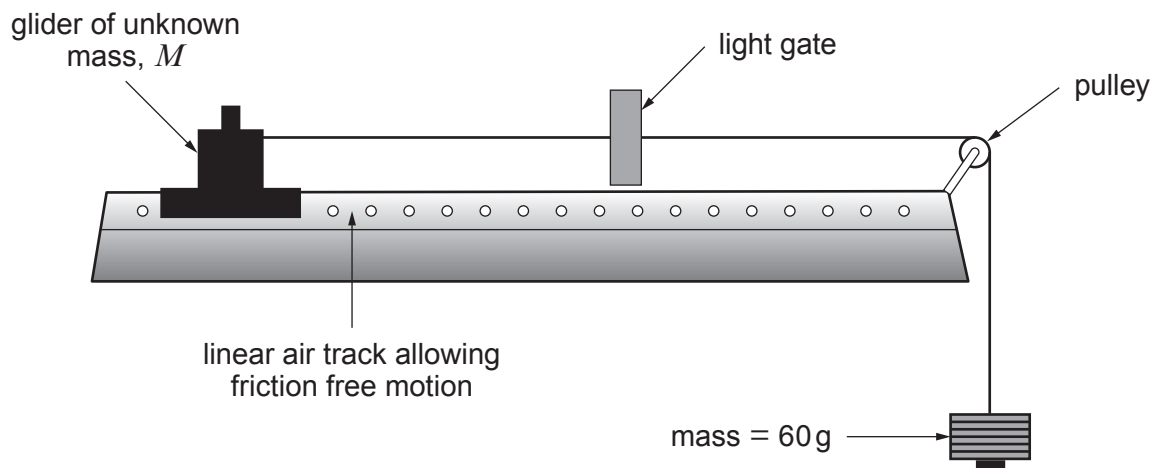


3. Emma uses the following apparatus to determine the unknown mass, M , of a glider.



Emma takes three readings of acceleration using a mass of 60g. She does this by attempting to hold the glider and releasing it from rest. The following values for acceleration are determined.

Trial	1	2	3
Acceleration / ms^{-2}	0.76	0.71	0.79

(a) (i) Calculate the mean acceleration along with the **absolute** uncertainty in its value. [2]

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(ii) Calculate the **percentage** uncertainty in the acceleration. [1]

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(b) The 60g mass consists of 6 slotted masses, each of $10.0\text{g} \pm 0.1\text{g}$.

(i) Determine the **percentage** uncertainty in the 60g mass. [1]

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- (ii) Hence determine the unknown mass, M , of the glider along with the **absolute** uncertainty in its value. Give both values to an appropriate number of significant figures. [5]

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- (c) Suggest how Emma could reduce the uncertainty in her value for M . [1]

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